



VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS):: BHIMAVARAM

approved by AICTE, accredited by NAAC-A+, NBA & affiliated to JNTUK

DEPARTMENT OF CSE (AI&DS)

III Year – II	L	T	P	C
Semester				
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BIG DATA ANALYTICS

Course objectives:

1. Optimize business decisions and create competitive advantage with Big Data analytics
2. Introducing Java concepts required for developing map reduce programs
3. Derive business benefit from unstructured data
4. Imparting the architectural concepts of Hadoop and introducing map reduce paradigm
5. To introduce programming tools PIG & HIVE in Hadoop ecosystem.

Course Outcomes:

At the end of the course students will be able to

1. Apply Java collections, generics, and serialization techniques in big data contexts.
2. Configure and work with Hadoop components and different cluster modes (local, pseudo, fully distributed).
3. Develop and execute MapReduce programs to process large datasets.
4. Analyze real-time and batch data using stream processing and Spark RDD operations.
5. Implement data analysis tasks using Pig Latin scripts and Hive queries for structured data.

UNIT-I:

Data structures in Java: Linked List, Stacks, Queues, Sets, Maps; Generics: Generic classes and Type parameters, Implementing Generic Types, Generic Methods, Wrapper Classes, Concept of Serialization

UNIT-II:

Working with Big Data: Google File System, **Hadoop Distributed File System (HDFS)**

Building blocks of Hadoop (Namenode, Datanode, Secondary Namenode, Job Tracker, Task Tracker), Introducing and Configuring Hadoop cluster (Local, Pseudo-distributed mode, Fully Distributed mode), Configuring XML files.

UNIT-III:

Writing Map Reduce Programs: A Weather Dataset, Understanding Hadoop API for Map Reduce Framework (Old and New), Basic programs of Hadoop Map Reduce: Driver code, Mapper code, Reducer code, Record Reader, Combiner, Practitioner

UNIT-IV:

Stream Memory and Spark: Introduction to Streams Concepts– Stream Data Model and Architecture, Stream computing, Sampling Data in a Stream, Filtering Streams, Counting Distinct Elements in a Stream, Introduction to Spark Concept, **Spark Architecture and components**, Spark installation, **Spark RDD (Resilient Distributed Dataset) – Spark RDD operations.**

UNIT-V:

Pig: Hadoop Programming Made Easier ,**Admiring the Pig Architecture**, Going with the Pig Latin Application Flow, Working through the ABCs of Pig Latin, Evaluating Local and Distributed Modes of Running Pig Scripts, Checking out the Pig Script Interfaces, Scripting with Pig Latin.

Applying Structure to Hadoop Data with Hive: Saying Hello to Hive, Seeing How the Hive Is Put Together, Getting Started with Apache Hive, Examining the Hive Clients, **Working with Hive Data Types, Creating and Managing Databases and Tables**, Seeing How the Hive Data Manipulation Language Works, Querying and Analysing data.

Text books:

- Wiley & Big Java 4th Edition, Cay Horstmann, Wiley John Sons, INC
- Hadoop: The Definitive Guide by Tom White, 3rd.Edition, O'reilly

Reference books:

- Hadoop in Action by Chuck Lam, MANNING Publ.
- Hadoop for Dummies by Dirk deRoos, Paul C.Zikopoulos, Roma B.Melnyk,Bruce Brown, Rafael Coss.
- Hadoop in Practice by Alex Holmes, MANNING Publ.
- Big Data Analytics by Dr. A.Krishna Mohan and Dr.E.Laxmi Lydia
- Hadoop Map Reduce Cookbook, SrinathPerera, ThilinaGunarathne